

NextGen Empirical Finance Research Camp — Week 8 Lesson Plan

Independent Research Project II — Execute · Stress-Test · Write · Defend

Prof. Lei Gao · George Mason University · Costello College of Business Session:

Friday, August 14, 2026 · 12:00-2:00 pm ET (Zoom).

— Where this session sits

This is the last live session, and it is the one every other week was preparation for. Students arrive with a **filed pre-analysis plan and identification memo** (the Week 7 deliverable) and a dataset in hand. Today they cross the finish line: **run the pre-registered test, try to break their own result on purpose, write it up to publication standard, and defend it out loud**. The capstone paper is due **August 15** — this session is the last supervised push before it.

The through-line of the whole camp lands here: a result you would defend to a skeptic. We built the discipline over seven weeks — now we execute it honestly, stress-test it without flinching, and present it without over-claiming. This maps to **textbook Week 8** (Ch 8.1–8.7, Lab 8). The mentor anchor is **Elnahas, Gao, Hossain & Kim (2024, *JFQA*)** — a published paper we read for what a referee actually asks.

— Learning objectives — by the end, a student can...

1. Run the **confirmatory** specification exactly as pre-registered, and keep an honest **deviations log**.
 2. Build and read a **specification curve** — the whole multiverse of reasonable specifications at once — and tell **robust** from **fragile**.
 3. Run a **robustness battery**: alternative standard errors, placebo/falsification tests, a wild-cluster bootstrap, and an Oster δ sensitivity check.
 4. Write the paper's **five-sentence skeleton**, a disciplined **250-word abstract**, and a **one-sentence contribution** — with causal language matched to the design.
 5. Assemble a **one-click replication packet**, take a **referee report**, and deliver an **8-minute defense** with three honest answers ready for Q&A.
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— Materials

- **Slides** — Week 8 GMU deck (24 slides, full speaker notes) + the **interactive web deck** (live specification-curve explorer, robustness battery, abstract-budget builder, referee lens).
 - **This lesson plan** and the **final assignment** — the Capstone Submission.
 - **Textbook Week 8** — Ch 8.1 (Execution & Specification Curve), 8.2 (Robustness), 8.3 (Writing), 8.4 (Peer Review), 8.5 (Presentation & Replication Packet), 8.6 (Talk/Poster/Defense), 8.7 (Submission); **Lab 8**; notebooks **nb8.1-nb8.5**.
 - **Anchors (all verified)** — Elnahas, Gao, Hossain & Kim (2024, JFQA) for the referee's questions; Rainbow of Municipal Credit, Gao & Sun (2019, PNAS), Deng, Gao & Kim (2020, JCF) for design; Simonsohn, Simmons & Nelson (2020) for the specification curve; Olken (2015) for pre-registration.
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Timed agenda (120 minutes)

0:00-0:10 — Welcome, recap & collect the deliverable (10 min)

Collect the Week 7 **Pre-Analysis Plan + Identification Memo**. Recap: the plan is filed; today it becomes a paper. Set the honest frame — whatever the result, effect or null, you report it and defend the design.

0:10-0:45 — Block A: Execute & stress-test (35 min)

1. **The confirmatory run.** Run the one pre-registered primary specification. Report it first, whatever it says. Any departure from the PAP goes in a **deviations log** — transparency, not erasure. 2. **The specification curve.** Your primary spec is one of hundreds of defensible choices (which controls, which sample, which fixed effects, which estimator). Plot the estimate across **all** of them at once — the multiverse. (Live demo on the interactive deck.) 3. **Reading it — robust vs. fragile.** A robust result holds its sign and significance across most of the curve. A fragile one lives or dies on a few lucky specifications. Report where on the curve your headline sits. 4. **The robustness battery.** Alternative standard errors, placebo/falsification tests, a **wild-cluster bootstrap** for few clusters, and an **Oster δ** check for how much unobserved selection it would take to kill the result. A result that survives tests capable of failing is one people believe.

0:45-0:50 — Break (5 min)

0:50-1:25 — Block B: Write it up to publication standard (35 min)

1. **The five-sentence skeleton.** Question → design → data → finding → contribution. If you can write those five sentences, you can write the paper. (Live builder on the interactive deck.) 2. **The 250-word abstract.** A hard budget forces clarity. (Live word-budget tool.) 3. **The one-sentence contribution.** "We are the first to show that ..." — and it must be true and supported. 4. **Causal-language discipline.** Match the verb to the design: "is associated with" for observational, "causes / raises / lowers" only when identification earns it. Over-claiming is the fastest way to lose a referee. 5. **Tables & figures to the Appendix-D standard**, and the **one-click replication packet** — a repo anyone can rerun with a single command to regenerate every number.

1:25-1:55 — Block C: Defend it (30 min)

1. **Peer review — the referee lens.** Read a claim like a referee: is the causal language calibrated, over-claimed, or under-claimed? Write the one hard question you would ask. (Interactive referee drill.) 2. **The 8-minute talk.** Decomposed: 1 min motivation, 2 min design, 2 min results, 1 min robustness, 1 min contribution, 1 min limits. Slides carry one idea each. 3. **The three honest answers for Q&A:** "I don't know — here's how I'd find out," "that's a limitation, and here's how I bounded it," "here's the exact evidence." **Mock defense round-robin:** each student gives their one-sentence contribution + takes one referee question from the room.

1:55-2:00 — Wrap & final assignment (5 min)

Assign the **Capstone Submission** — the completed paper + one-click replication packet + 8-minute presentation + AI-use disclosure — **due August 15** (presentation/conference day). This is the program's **terminal deliverable**. Close on the long arc: today's poster is the first

— Instructor notes

- **Protect Block C.** The mock defense is where students learn their result is theirs to defend — keep it to crisp one-question exchanges.
- **Demo the specification curve live.** The interactive deck's explorer makes "researcher degrees of freedom" visceral in a way a static slide cannot — toggle a control on and off and watch the headline move.
- **Label all illustrative numbers.** The interactive widgets use clearly-marked **simulated** data to demonstrate method — never present them as real estimates.
- **Discipline the causal verbs.** The single most common capstone error is a design that identifies an association paired with a sentence that claims causation. Catch it here.
- **Verified anchors only.** Elnahas, Gao, Hossain & Kim (2024, JFQA); Rainbow of Municipal Credit; Gao & Sun (2019); Deng, Gao & Kim (2020) — model the citation discipline the whole camp taught.